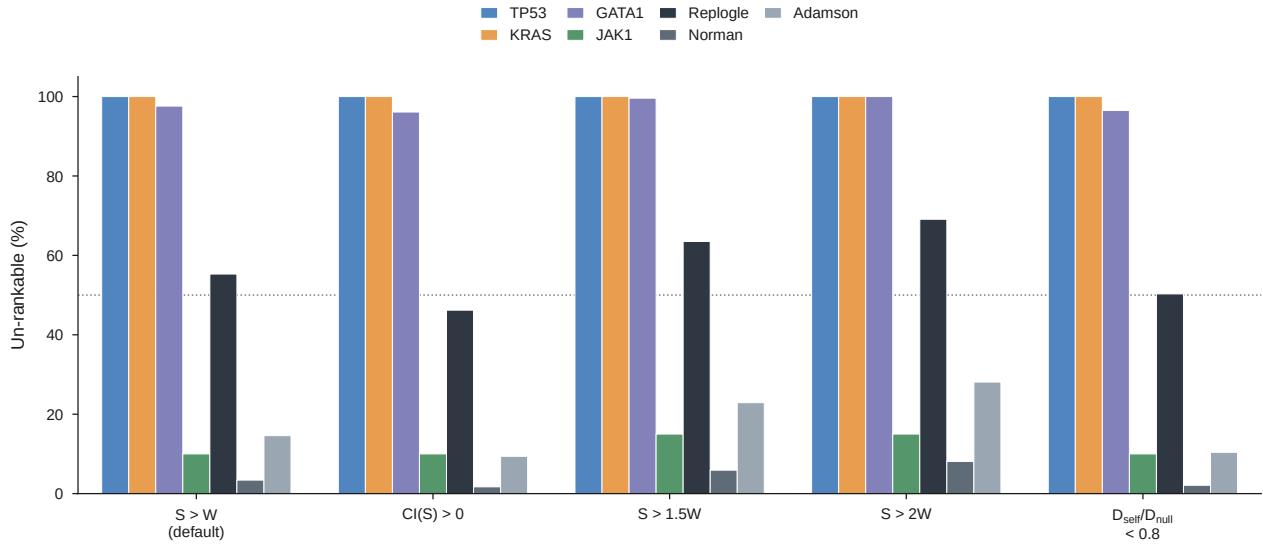


Supplementary Information

Measurement and model limits of allele-specific single-cell perturbation prediction

This document contains the full parameter tables underlying the variant feature representation, the prediction methods and the split-half rankability analysis. Complete assignments and code are also available at <https://github.com/Boom5426/AllelePerturb>.



Supplementary Figure 1: **The rankability verdict is robust to the choice of criterion.** Fraction of un-rankable perturbations for the seven datasets (the four AllelePerturb genes and three gene-level Perturb-seq atlases) under five rankability criteria: the primary signal-versus-noise rule $S > W$ (default); a bootstrap lower-bound rule ($CI(S) > 0$, the lower 95% confidence bound of S above zero); two increased-stringency rules ($S > 1.5W$ and $S > 2W$); and a ratio threshold, under which a variant is un-rankable when $D_{self}/D_{null} \geq 0.8$. Absolute fractions shift with stringency, but the ordering across datasets is preserved under every criterion: TP53 and KRAS remain $\sim 100\%$ un-rankable, GATA1 stays high, JAK1 stays low ($\sim 10\%$), and among gene-level atlases Replogle is consistently the highest and Norman the lowest. Source values are listed in `results/rankability_sensitivity.csv`.

Supplementary Table 1: **Variant feature (θ) structural and hotspot annotations.** Per-gene residue assignments for the three structural components of the biophysical feature vector. The fold-core term f_v was additionally set to 0.5 for annotated interaction regions and 0.0 elsewhere. The hotspot/pathogenic marker h_v was defined exclusively from external prior knowledge, with no reference to the single-cell outcome.

Gene	Structural core ($f_v = 1$)	Catalytic switch ($s_v = 1$)	/	Hotspot / pathogenic ($h_v = 1$)
TP53	DNA-binding domain, residues 94–293	–		COSMIC/IARC missense hotspot codons 175, 176, 179, 220, 238, 245, 248, 249, 273, 282, 285
KRAS	G-domain, residues 1–169	G12, G13, Q61		COSMIC/IARC missense hotspot codons 12, 13, 59, 61, 117, 146
GATA1	Zinc fingers (204–228) and ZF2 (258–282)	ZF1 Zinc-finger cys-teines		Zinc-coordinating cysteines 204, 207, 225, 228, 258, 261, 279, 282; ClinVar pathogenic / likely-pathogenic annotations
JAK1	Kinase domain, residues 866–1154	–		Functional-module membership: JH2 pseudokinase (583–855), JH1 kinase (875–1153)

Note. An earlier version of h_v used the top quintile of measured effect size, which leaks the evaluation outcome into the feature; it was removed and the entire evaluation grid was re-run, with ranking performance unchanged (θ -only PDS shifted by at most 0.02 and every method remained at chance).

Supplementary Table 2: **Implementation and sampling parameters.** *Regression heads (top):* the six heads were crossed with three feature spaces (θ , ESM, ESM+ θ) to form the 18 in-house feature-model combinations; all heads used a fixed random seed and were fit by minimizing squared error on the training variants. *Split-half and power-analysis sampling grid (bottom):* parameters for the energy-distance split-half analysis (Fig. 4) and the depth power analysis (Figs. 4, 5), each configuration summarized by the median and the 2.5–97.5 percentile interval across seeds.

Head	Family	Settings
Ridge regression	Linear (L2)	$\alpha = 1.0$
Lasso regression	Linear (L1)	$\alpha = 0.01$; up to 2,000 iterations
Random forest	Tree ensemble	50 trees; maximum depth 6
Gradient-boosted trees	Tree ensemble	50 estimators; maximum depth 3; multi-output wrapper
k -nearest-neighbors	Neighbor	$k = 3$
Multilayer perceptron	Neural network	hidden layers 128 and 64; ReLU; up to 500 iterations

Sampling parameter	Values
Split-half subsample sizes (per half)	25, 50, 100, 150, 250, 400 cells
Representation spaces	50-component PCA (canonical), gene space, pathway-score space
Random seeds per configuration	50
Power-analysis depths (per variant)	50, 100, 150, 300 cells
Minimum cells for inclusion	50 (split-half); 5 (pseudobulk profiles)

Note. Linear and neighbor heads regress directly onto the full expression profile. Because tree ensembles scale poorly to thousands of output genes, for output spaces larger than 200 genes the random forest and gradient-boosted heads were fit on a q -component PCA of the training profiles and predictions were mapped back to gene space by the inverse transform, with $q = \min(50, G, n_{\text{train}} - 1)$.

Supplementary Table 3: **Detection, identification and prediction: a decision table for allele-specific single-cell perturbation prediction.** Three nested questions of increasing difficulty locate a dataset by its failure mode: detection (a variant versus wild type; assay sensitivity), identification (one variant versus another; ground-truth resolution) and prediction (a held-out variant from its features; computational generalization). Prediction is only meaningful once identification is achievable. Detection is graded rather than binary: in the three datasets with low measured effect size the variant-to-wild-type separation is marginal (near the replicate-noise floor; Fig. 4c,e), so ‘marginal’ detection denotes a weak but non-zero separation from wild type. Marginal identification likewise denotes a small fraction of resolvable sibling pairs with no reliable recovery of a variant’s nearest sibling. Because each gene here is also a distinct dataset, assay and cell context, a row locates a measurement and modelling regime rather than an intrinsic property of the gene.

Detection	Identification	Prediction	Limiting regime	Dataset in this study
no	–	–	assay-limited	(none)
marginal	no	not evaluable	measurement-limited	TP53, KRAS
marginal	marginal	not evaluable	measurement-limited	GATA1
yes	yes	no	model-limited	JAK1
yes	yes	yes	successful regime	(target)

Supplementary Table 4: **Published perturbation models under a unified allele-level evaluation.** Perturbation-discrimination score (PDS, cosine) with 95% bootstrap confidence intervals, pooled over five generalization splits and four genes; chance is 0.50. Variant-conditionable models are scored directly; gene-keyed models are allele-blind by construction and reported by resolution coverage. Every method was scored through one harness (identical held-out variants, pseudobulk profiles, tie-aware PDS and permutation null).

Model family	Method	PDS (95% CI)	Pearson- δ
<i>Variant-conditionable (run to completion)</i>			
Optimal-transport flow	CellFlow	0.51 (0.48–0.54)	0.12
Attribute disentangle- ment	Biolord	0.49 (0.47–0.52)	0.13
Latent arithmetic (dose)	scVIDR	0.50 (0.48–0.53)	0.13
Latent arithmetic	scGen	0.49 (0.46–0.52)	0.18
Conditional-flow gener- ative	PerturbNet [†]	0.54 (0.51–0.57)	0.13
Feature regression	baselines (Fig. 2)	0.49–0.52	0.55–0.65
<i>References</i>			
Gene identity	Gene-mean	0.49 (0.46–0.52)	0.66
No perturbation	WT-null	0.50	–
<i>Allele-blind by construction (resolution coverage 4/470 = 0.85%)</i>			
Foundation model	scGPT	n/a [‡]	–
Gene knowledge graph	GEARS	n/a [‡]	–
Virtual-cell (one-hot)	STATE	n/a [‡]	–
<i>Did not converge</i>			
Compositional autoen- coder	variant-CPA	non-finite loss	–

[†]PerturbNet generates within a 50-dimensional principal-component subspace in which random predictions already score 0.52; full-space-at-chance predictors (Ridge-esm, Gene-mean) projected into the same subspace are likewise inflated to 0.52, and its exceedance over a subspace permutation null is not significant (permutation $P = 0.13$). [‡]These models condition on perturbed-gene identity and assign an identical prediction to every variant of a gene, so an allele-level PDS is undefined (a category error).

Supplementary Table 5: **Per-gene detection ratio, achievable ceiling and model performance.** Split-half energy-distance ratio $D_{\text{self}}/D_{\text{null}}$; oracle discrimination ceiling (independent second measurement, estimating attainable discrimination); fraction of allele pairs resolvable above replicate noise; fraction of variants rankable at native depth; and best in-house model PDS (canonical multi-seed harness). Chance is 0.50.

Gene	$D_{\text{self}}/D_{\text{null}}$	Oracle ceiling (95% CI)	Pairs resolvable	Rankable (native)	Best model PDS [†]
TP53	0.96	0.485 (0.445–0.525)	0%	0%	0.51
KRAS	1.00	0.500 (0.440–0.566)	0%	0%	0.54
GATA1	0.88	0.572 (0.549–0.595)	3.8%	2.4%	0.50
JAK1	0.21	0.792 (0.742–0.838)	78.5%	90%	0.52

[†]Best model is the highest-scoring in-house feature-regression head per gene under the canonical multi-seed harness (published variant-conditionable models are compared separately in Supplementary Table 4). PerturbNet is excluded; its higher per-gene values (0.53–0.54) arise from its 50-dimensional principal-component subspace, in which random predictions already score 0.52.

Supplementary Table 6: **Per-profile sampling noise and between-variant signal, by gene and depth.** Each variant's cells are cut into four disjoint groups of m cells: two score the replicate ceiling and two estimate the signal, so no noise realisation enters both. $\bar{\Delta}^2$ is the mean squared separation between variants after subtracting the sampling-noise term, reported unclipped with a 95% interval bootstrapped over both variants and cell splits; it is negative about half the time when the true separation is at the floor, and clipping it at zero would report certainty the data do not carry. P is a permutation test re-partitioning the half-profiles among pseudo-variants, and is conservative because the null averages five splits and the observed estimate twenty. Rows are the eligible cohort, every variant with $4m$ cells; the file also carries a fixed cohort held constant across depths, which is underpowered and is not reproduced here. Source: results/canonical/floor_low_v2.csv.

Gene	m	n	η^2	$\bar{\Delta}^2$ (95% CI)	ρ^2	P	Ceiling
TP53	25	98	37.61	-0.051 (-0.403, +0.339)	-0.0007	0.581	0.499
TP53	50	98	18.87	-0.034 (-0.231, +0.156)	-0.0009	0.630	0.499
TP53	100	68	9.39	-0.050 (-0.162, +0.069)	-0.0026	0.786	0.497
TP53	150	35	6.29	-0.060 (-0.164, +0.036)	-0.0048	0.840	0.473
KRAS	25	92	37.57	-0.095 (-0.400, +0.210)	-0.0013	0.686	0.495
KRAS	50	92	18.78	-0.066 (-0.247, +0.108)	-0.0017	0.757	0.497
KRAS	100	84	9.39	-0.022 (-0.138, +0.096)	-0.0011	0.638	0.498
KRAS	150	49	6.25	+0.055 (-0.043, +0.150)	+0.0044	0.135	0.510
GATA1	25	247	98.18	+2.373 (+0.561, +4.170)	+0.0121	0.009	0.512
GATA1	50	209	49.31	+2.153 (+0.766, +3.670)	+0.0218	0.001	0.524
GATA1	100	107	24.82	+0.605 (-0.226, +1.486)	+0.0122	0.072	0.520
GATA1	150	57	16.46	+0.778 (-0.056, +1.736)	+0.0236	0.021	0.521
GATA1	250	15	9.67	+0.865 (+0.128, +1.700)	+0.0447	0.035	0.540
JAK1	25	14	78.11	+97.183 (+56.173, +126.186)	+0.6221	0.001	0.780
JAK1	50	9	39.92	+83.626 (+8.442, +140.473)	+1.0475	0.001	0.821
JAK1	100	5	19.74	+128.768 (+4.291, +205.959)	+3.2615	0.002	0.970

Supplementary Table 7: **Depth required to reach a target replicate ceiling, and where the calculator refuses.** The nearest-competitor separation belongs to the perturbations and does not change with depth while the noise falls as $1/m$, so resolution scales linearly from a pilot and the depth reaching a target is read off the calibration by inversion. The calculator returns nothing when the pilot separation's lower confidence bound does not clear zero: no finite depth recovers a separation that is not there, and a number extrapolated from noise reads as a plan. JAK1 is a check on the scaling rather than a fit, since three pilots at different depths, on different cells and different candidate sets, agree within a factor of two. Source: results/canonical/resolution_low_prescriptions.csv.

Gene	Pilot	n	Nearest-competitor ρ^2 (95% lower)	Cells per group to reach	
	depth			ceiling 0.7	ceiling 0.9
TP53	25	98	-0.0026 (-0.0058)	refused	refused
TP53	50	98	+0.0005 (-0.0079)	refused	refused
TP53	100	68	+0.0019 (-0.0033)	refused	refused
KRAS	25	92	+0.0004 (-0.0056)	refused	refused
KRAS	50	92	+0.0009 (-0.0063)	refused	refused
KRAS	100	84	-0.0073 (-0.0133)	refused	refused
GATA1	25	247	+0.0157 (+0.0027)	62	232
GATA1	50	209	+0.0010 (-0.0108)	refused	refused
GATA1	100	107	+0.0051 (-0.0141)	refused	refused
JAK1	25	14	+0.0750 (+0.0486)	10	55
JAK1	50	9	+0.1273 (+0.0910)	12	65
JAK1	100	5	+0.1670 (+0.1419)	19	99

Supplementary Table 8: **Every method on both scoring axes, each against its own permutation null.** The residual axis scores after removing a gene-shared programme computed from training variants only. Its chance level is not 0.50: WT-null, which predicts a zero vector and therefore one constant shared by every held-out variant, scores 0.524 and its permutation null is 0.524 exactly, because training variants entering the mean have their residuals shrunk toward the origin while held-out variants do not. Nulls are formed by permuting predictions across the held-out variants of a gene; intervals are 2,000 bootstraps over variants. Six methods (*) exceed their own null at $P < 0.05$, by 0.011 to 0.025. None is established: fifty permutations floor the attainable P at 0.0196, four of the six sit on that floor, and twenty-five methods were tested. Source: results/canonical/residual_axis_models.csv.

Method	n	PDS (95% CI)	null	residual-PDS (95% CI)	null	P
scVIDR	308	0.540 (0.505–0.574)	0.532	0.557 (0.522–0.590)	0.552	0.118
KNN-theta	308	0.486 (0.456–0.518)	0.473	0.555 (0.526–0.584)	0.531	0.118
KNN-esm+theta	305	0.486 (0.456–0.516)	0.477	0.554 (0.524–0.580)	0.531	0.118
KNN-esm	305	0.483 (0.452–0.514)	0.478	0.540 (0.511–0.570)	0.534	0.333
PerturbNet	303	0.537 (0.510–0.567)	0.519	0.539 (0.510–0.571)	0.529	0.216
MLP-theta*	308	0.455 (0.426–0.485)	0.449	0.538 (0.507–0.567)	0.513	0.020
scGen*	308	0.483 (0.452–0.515)	0.482	0.528 (0.498–0.558)	0.517	0.020
GBoost-theta*	308	0.475 (0.447–0.505)	0.464	0.528 (0.501–0.555)	0.512	0.039
RF-theta*	308	0.455 (0.425–0.485)	0.451	0.527 (0.498–0.554)	0.512	0.039
Ridge-theta*	308	0.460 (0.431–0.489)	0.449	0.525 (0.498–0.552)	0.502	0.020
CellFlow	303	0.501 (0.471–0.532)	0.501	0.524 (0.493–0.554)	0.524	0.471
WT-null	308	0.500 (0.500–0.500)	0.500	0.524 (0.493–0.554)	0.524	1.000
GBoost-esm+theta	305	0.471 (0.442–0.500)	0.462	0.523 (0.498–0.548)	0.508	0.078
RF-esm+theta	305	0.454 (0.424–0.484)	0.452	0.522 (0.495–0.547)	0.512	0.196
Biolord*	308	0.497 (0.467–0.528)	0.493	0.521 (0.491–0.551)	0.504	0.020
Ridge-esm+theta	305	0.461 (0.432–0.491)	0.454	0.521 (0.493–0.548)	0.505	0.059
GBoost-esm	305	0.471 (0.442–0.499)	0.463	0.516 (0.491–0.541)	0.508	0.255
RF-esm	305	0.452 (0.422–0.482)	0.451	0.516 (0.490–0.543)	0.511	0.372
Ridge-esm	305	0.451 (0.421–0.481)	0.453	0.512 (0.487–0.538)	0.505	0.353
Lasso-theta	308	0.458 (0.427–0.487)	0.449	0.510 (0.484–0.536)	0.500	0.157
Lasso-esm+theta	305	0.452 (0.421–0.482)	0.451	0.506 (0.479–0.533)	0.500	0.275
MLP-esm	305	0.449 (0.419–0.479)	0.450	0.503 (0.473–0.532)	0.503	0.529
Gene-mean	308	0.447 (0.417–0.476)	0.447	0.500 (0.500–0.500)	0.500	1.000
MLP-esm+theta	305	0.448 (0.417–0.478)	0.449	0.499 (0.469–0.530)	0.492	0.176
Lasso-esm	305	0.450 (0.420–0.479)	0.451	0.493 (0.463–0.524)	0.495	0.804

Supplementary Table 9: **The frozen criterion applied to six published perturbation screens.** The criterion, its parameters, the datasets and four predictions were fixed before any dataset was measured (docs/PREREG_RESOLUTION_PANEL_v1.md); nothing was tuned afterwards and no dataset was dropped. Three of the six took no part in developing the criterion. Not one is benchmarkable: every screen has a far lower identifiable than detectable fraction, and the two largest atlases still order graded predictors at 0.997 and 0.96 while 0.5% and 1.4% of their perturbations are identifiable. One of the four predictions failed: Adamson and Replogle are reversed on the detectable fraction relative to the manuscript’s own pipeline, by 3.3 percentage points, so the packaged criterion and that pipeline are not interchangeable. Source: results/canonical/resolution_panel.csv.

Dataset	n	Detectable (%)	Identifiable (%)	Ceiling	$P(\text{order})$	Verdict
McFarland	10	90.0	20.0	0.928	0.418	detectable
Tahoe-100M	7	71.4	42.9	0.929	0.106	detectable
Norman 2019	195	68.7	0.5	0.898	0.997	detectable
Replogle 2022	414	51.2	1.4	0.865	0.959	detectable
Adamson 2016	94	47.9	3.2	0.851	0.974	not detectable
Virtual Cell Challenge	139	25.9	4.3	0.806	0.903	not detectable

Supplementary Note: robustness and sensitivity controls.

Statistical independence of variants. Because some residues carry more than one substitution, the 470 variants are not fully independent. Under the random split, 47% of held-out variants share a residue with a training variant (positional 0%, mechanistic 10%, low-depth 37%, compatibility 54%); the random split is therefore an optimistic setting, yet allele discrimination remained at chance under it and under the residue-disjoint positional split alike. Resampling residue-level clusters rather than individual variants in the bootstrap left the per-method PDS intervals essentially unchanged: all 23 non-null predictors continued to overlap the 0.50 chance level (only PerturbNet excluded it, the subspace artifact described above), and Pearson- δ remained above zero for every feature-model head. The direction-versus-ranking dissociation is therefore not an artifact of treating correlated substitutions as independent units. Source: results/reviewer_controls/residue_independence_summary.csv.

Wild-type reference construction. To test whether the detection floor reflects the variant effect or the heterogeneity of the wild-type reference, we compared, at matched depth, the variant replicate distance D_{self} , the variant-to-wild-type distance D_{null} and the wild-type-to-wild-type distance D_{wtwt} from an independent split of the wild-type cells (energy distances, 50-component PCA space). For TP53, KRAS and GATA1 the three distances coincided ($D_{\text{wtwt}}/D_{\text{null}} = 0.91\text{--}1.09$ across 50–300 cells per half), so a variant differs from wild type no more than wild type differs from itself, and D_{self} , which involves no wild-type cells, sits at the same level: the floor is set by the absence of variant signal relative to sampling noise, not by an inflated or heterogeneous control. JAK1 shows the contrast, with D_{null} eight- to thirty-fold above D_{wtwt} ($D_{\text{wtwt}}/D_{\text{null}} = 0.03\text{--}0.12$). The ratios are stable across wild-type sampling depth, so the floor is not a depth artifact. Source: results/reviewer_controls/wt_control_diagnostic.csv.

Metric-family invariance. The detection floor is not specific to energy distance. The split-half window ratio $D_{\text{self}}/D_{\text{null}}$ (50-component PCA, 50 cells per half) stayed near one for TP53, KRAS and GATA1 and low for JAK1 under all five distance families tested (energy, RBF maximum-mean-discrepancy, sliced-Wasserstein, cosine and Euclidean): TP53 1.01–1.03, KRAS 0.95–0.99, GATA1 0.90–1.04, JAK1 0.05–0.40. Source: results/canonical/metric_family_floor.csv.

Gene-shared versus allele-specific decomposition. Splitting each effect as $\delta_v = \bar{\delta}_g + r_v$ apportions the measured response between a component shared by all variants of a gene and an allele-specific residual; the shared fraction is a property of each dataset rather than of the metric, at 72% (TP53), 59% (KRAS), 29% (GATA1) and 51% (JAK1). What predictors actually recover follows instead from scoring the predictions against the residual: across all 24 predictors the residual Pearson- δ lay between -0.08 and 0.03 , so conventional direction recovery is carried almost entirely by the gene-shared component. An oracle scored on the allele-specific residual profiles, with the gene mean taken from training variants only, stayed at the axis’s own 0.524 reference for TP53 and KRAS (0.509 and 0.518), was marginal for GATA1 (0.576) and high for JAK1 (0.839), so the allele-specific residual is reproducible only where the measurement resolves sibling variants. An earlier decomposition centred on a mean that included the variants being scored and gave 0.892 for JAK1; self-inclusion in the mean makes that figure optimistic. Source: `results/canonical/residual_axis_ceiling.csv`.

Representation robustness. Under identical splits, regression heads and scoring (single deterministic pseudobulk draw; the multi-seed convention used in the main text gives PDS 0.49–0.52), every variant representation left allele discrimination at chance while recovering direction similarly: PDS-cosine 0.458–0.473 and Pearson- δ 0.628–0.659 for the biophysical θ vector, ESM-1v, four ESM2 constructions (global, mutant-minus-wild-type global, mutated-residue and local-window) and AlphaMissense; a featureless training-gene-mean predictor reached Pearson- δ 0.66 at PDS 0.44. The scoring procedure is identical across representations, but variant coverage is not, because each representation is defined only where its input exists: AlphaMissense scores were available for 374 of the 470 benchmark variants (84 of 98 for TP53, 78 of 92 for KRAS, 201 of 254 for GATA1 and 11 of 26 for JAK1), giving 389 scored variant-by-split instances against 518 for θ , 511 for ESM-1v and 497 for the ESM2 constructions. Correlations between AlphaMissense pathogenicity and measured effect size are therefore reported per gene only (Spearman -0.08 for TP53, -0.05 for KRAS and 0.09 for GATA1); cross-gene pooling is not interpretable here, because both the pathogenicity distribution and the scale of the measured transcriptional effect differ between genes. Sources: `results/reviewer_controls/rep_grid_summary.csv`, `am_summary.csv` and `am_effect_spearman.csv`.

Oracle ceiling sensitivity. The replicate ceiling was unchanged by depth (300-cell versus full-depth) and stayed at chance for TP53 and KRAS even for a perfect prediction scored against the benchmark’s noisy pseudobulk truth (TP53 0.488, KRAS 0.490); GATA1 remained marginal (0.569) and JAK1 high. Shrinking a prediction fully to the gene mean lowered discrimination to 0.500 for every gene, so mean-shrinkage denoising does not raise the ceiling.

Train-only preprocessing. Refitting the per-gene standardization on training variants and wild-type cells only (excluding held-out variants) left every gene at or below chance: per-gene PDS-cosine, all-cell versus train-only, was 0.47/0.44 (TP53), 0.49/0.50 (KRAS), 0.46/0.46 (GATA1) and 0.46/0.45 (JAK1), with direction recovery essentially unchanged (mean Δ PDS = -0.01), so the reported all-cell standardization is conservative with respect to the null. Source: `results/canonical/leakage_control_summary.csv`.